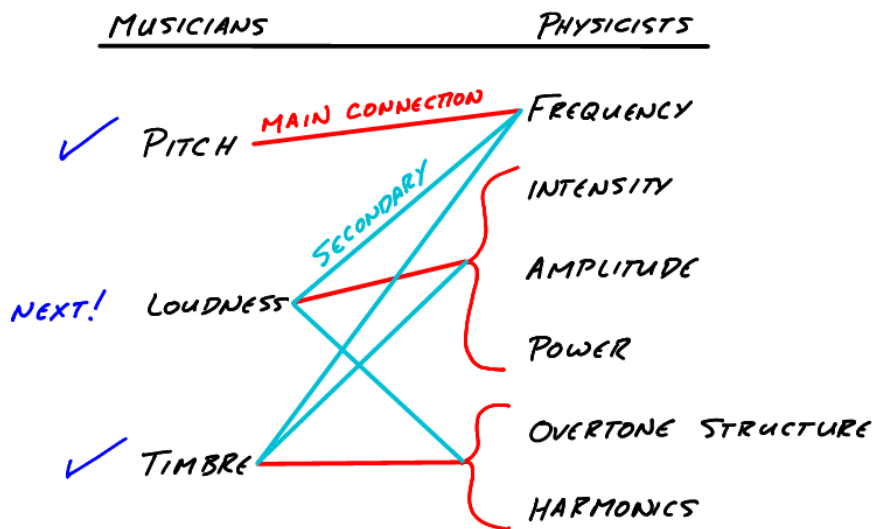


LOUDNESS

PROGRESS:



OUTLINE:

- ENERGY BASICS
 - ENERGY IN WAVES
 - POWER
 - INTENSITY
 - DECIBEL (LOG) SCALE
 - EAR
 - LOUDNESS
- } BUILDING BLOCKS
- } 2 WAYS TO QUANTIFY THE SAME THING

END OF SEPT 16

ENERGY "STUFF THAT CAN MAKE STUFF MOVE"

INTUITIONS:

- MOVING OR "KINETIC" ENERGY:

FASTER = MORE
HEAVIER = MORE } WRECKING BALLS ARE NOT
MOTIONLESS STYROFOAM

- STORED OR "POTENTIAL" ENERGY

HIGHER, HEAVIER OBJECTS (IN GRAVITY) HAVE MORE.
MORE STRAINED, STIFFER SPRINGS HAVE MORE.

↖ DROP A
WRECKING
BALL FROM
HIGHER!

↖ ADD YOUR
NEEF GUNS,
FOLKS.

ENERGY COMES IN MORE FORMS THAN WE WILL DISCUSS:

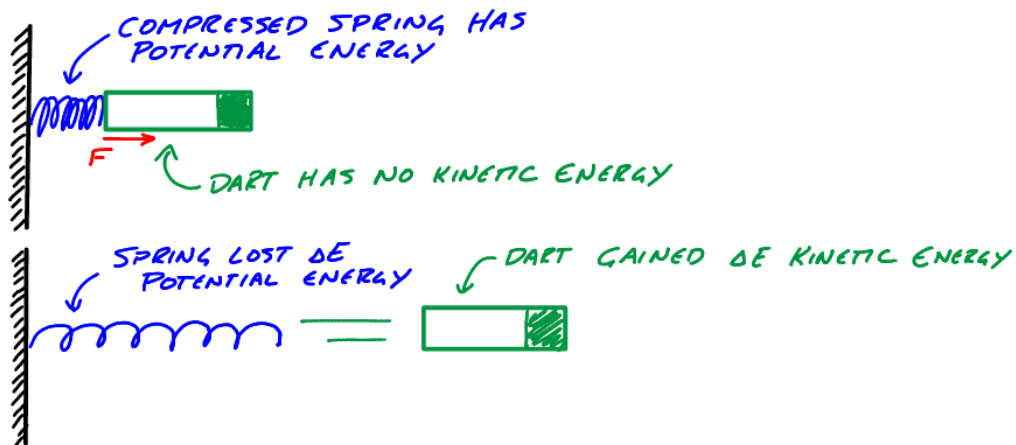
- KINETIC
- GRAVITATIONAL
- MATERIAL STRAIN
- ELECTROMAGNETIC
- CHEMICAL
- NUCLEAR
- HEAT
- MATTER ITSELF (!)

WORK: IF YOU PUSH WITH FORCE F ON SOMETHING FOR A DISTANCE Δx
YOU TRANSFER ENERGY

$$\Delta E = F \Delta x \quad \text{"WORK" DONE BY A FORCE.}$$

TO IT (AND YOU LOSE ΔE)

- EASIEST CASE: SPRING PUSHING A DART



- NOTE IF F IS POSITIVE + Δx IS NEGATIVE, $\Delta E < 0 \Rightarrow$ ENERGY IS TRANSFERRED THE OTHER WAY. (IMAGINE THE DART FLYING

THE OTHER WAY + COMPRESSING THE SPRING AS IT STOPS)

- NOTE WHEN YOU TRY TO STOP A ROLLING CAR, YOU DON'T STORE THE ENERGY GIVEN TO YOU. YOU JUST CONVERT IT TO HEAT, NOISE, SHOE (UNLESS IT SENDS YOU ☺) + TISSUE DAMAGE...

- NOTE YOU ALSO DAMAGE TISSUE + GENERATE HEAT + NOISE, ETC, WHEN PUSHING ON SOMETHING THAT DOESN'T MOVE.

ENERGY ALWAYS GOES SOMEWHERE!

KINETIC ENERGY

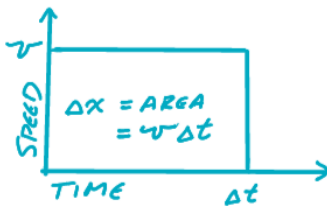
- SUPPOSE WE APPLY FORCE F TO A MASS m (IN SPACE, STARTING AT REST)
RECALL IT WILL ACCELERATE AS

$$a = \frac{F}{m}$$

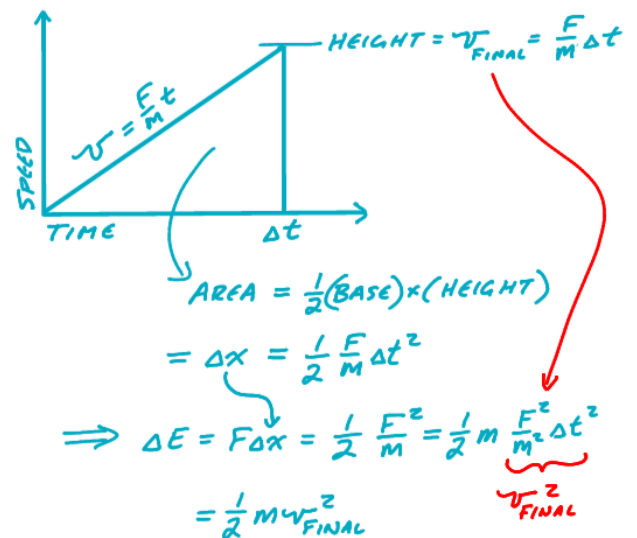
WHICH MEANS ITS SPEED WILL BE $v(t) = at = \frac{F}{m}t$

BABY CALCULUS BREAK:

- YOU ALL BELIEVE (DISTANCE) IS (SPEED) $\times$ (TIME)
- GRAPHICALLY, THIS IS THE AREA OF A RECTANGLE



- IF v IS CHANGING, THIS IS NO DIFFERENT! (BELIEVE THIS OR ASK ME WHY :))



HEAVIER, FASTER $\Rightarrow$ MORE KINETIC ENERGY
DOES NOT DEPEND ON DIRECTION

$$\Rightarrow E_{\text{KINETIC}} = \frac{1}{2} m v^2$$

KINETIC ENERGY, UNITS $\text{Nm} = \text{J}$
"JOULE"

2's COME FROM AREA OF TRIANGLE CALCULATION

- IF YOU INTERNALIZE THE IDEA OF Δx BEING THE AREA UNDER THE $v(t)$ LINE, YOU CAN SOLVE MOST INTRO PHYSICS PROBLEMS WITH ALMOST NO EQUATIONS MEMORIZED. IT'S ALL JUST RECTANGLES + TRIANGLES.

END OF W60
SEPT 18 (QUIZ)



(I ALSO USED THIS EXAMPLE TO LEARN MOST OF CALCULUS)

POTENTIAL ENERGY

- MOVING AN OBJECT AGAINST A FORCE SLOWS IT DOWN, AND THE LOST KINETIC ENERGY CAN SOMETIMES BE STORED.

EXCLUDING HEAT, SOUND, OR OTHER LOSSES

- EXAMPLES:

- EARTH'S GRAVITY PULLS DOWNWARD WITH FORCE

$$F_{\text{GRAV}} = m \overset{\text{MASS}}{g} \leftarrow \text{ACCELERATION CONSTANT} \approx 9.8 \text{ m/s}^2$$

LIFTING A MASS m AGAINST GRAVITY A DISTANCE Δx

$$\Rightarrow \text{WORK } \Delta E = \boxed{mg\Delta x \equiv U_{\text{GRAVITY}}} \quad \begin{array}{l} \text{GRAVITATIONAL POTENTIAL} \\ \text{(EARTH'S SURFACE)} \end{array}$$

$\uparrow$ HEIGHT

THIS IS A POTENTIAL ENERGY BECAUSE, IF DROPPED, THE MASS WILL GAIN THIS MUCH KINETIC ENERGY AFTER TRAVELLING Δx DOWNWARD. IT IS THE WORK THAT "CAN BE" DONE.

$$\Rightarrow U_{\text{GRAVITY}} = mg\Delta x \rightarrow E_{\text{KINETIC}} = \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = 2g\Delta x \quad \left(\text{FAMOUS "ALL MASSES FALL THE SAME"} \right)$$

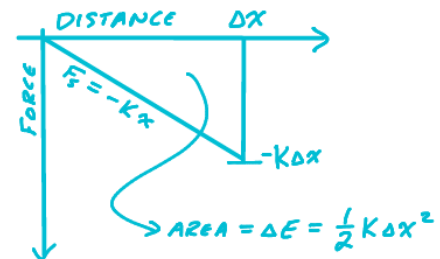
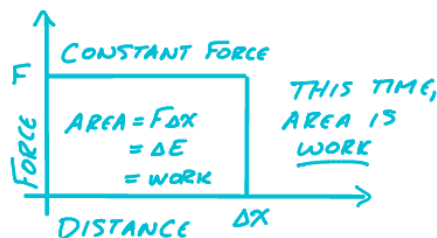
- SPRING:

- WHEN COMPRESSING OR EXTENDING A SPRING FROM ITS EQUILIBRIUM LENGTH, IT EXERTS FORCE

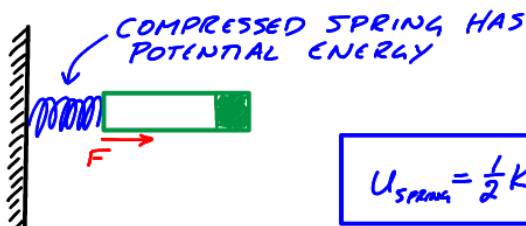
$$F_s = -K\Delta x$$

$\leftarrow$ SPRING CONSTANT, UNITS N/M (NOT BOLTZMANN k_B)
MORE COMPRESSED = MORE FORCE.

BABY CALCULUS 2:



- NOW BE CAREFUL ABOUT THE SIGN: IF I PUSH AGAINST THE SPRING'S FORCE, I GIVE THE SPRING POTENTIAL ENERGY



$$\boxed{U_{\text{SPRING}} = \frac{1}{2}K\Delta x^2} \quad \text{SPRING POTENTIAL ENERGY}$$

- SIMILAR TO GRAVITY, THIS ENERGY CAN ACCELERATE A MASS TO SPEED v , BUT NOW ACCORDING TO

$$\frac{1}{2}K\Delta x^2 = \frac{1}{2}mv^2$$

10pt
Q

DOES A STRETCHED SPRING HAVE POTENTIAL ENERGY?
YES! (SLINGSHOT?)

- WITH THOSE "BASICS" IN HAND, LET'S DISCUSS ENERGY CARRIED BY WAVES:

WAVE ENERGY DENSITY



- SO HOW MUCH ENERGY DOES A WAVE HAVE?

- WAVES HAVE REGIONS OF POTENTIAL + KINETIC ENERGY
(STRETCHED) (MOVING)

10pt

Q FOR THE STRING WAVE BELOW, WHERE IS THE MAXIMUM POTENTIAL ENERGY (DENSITY)? KINETIC?



DEMO PHET: WAVE ON A STRING

- NO END
- NO DAMPING
- SLOW
- OSCILLATE
- NOTE STRETCH + BALL SPEEDS.

- CHANGE AMPLITUDE + NOTICE THE AMOUNT OF STRETCH AND VELOCITY SCALES WITH IT.

$\Rightarrow \Delta x \propto \text{AMPLITUDE}$ FOR EACH BALL
 $v \propto \text{AMPLITUDE}$ FOR EACH BALL

$\Rightarrow U_{\text{spring}} = \frac{1}{2} K \Delta x^2 \propto (\text{AMPLITUDE})^2$

$E_{\text{kinetic}} = \frac{1}{2} m v^2 \propto (\text{AMPLITUDE})^2$

$\Rightarrow \text{WAVE ENERGY DENSITY} \propto (\text{AMPLITUDE})^2$

CAN USE "PHYSICS" TO FIND THE PROPORTIONALITY CONSTANT, OR JUST PAY ATTENTION TO SCALING FOR THIS COURSE.

- TRUE FOR ALL WAVES I HAVE SEEN

- STRING [J/m]
- MEMBRANES [J/m²]
- SOUND [J/m³]

AIR CAN BEHAVE LIKE A SPRING TOO!

HONESTLY, THOUGH, WE LIVE IN 3D, SO THEY'RE ALL ACTUALLY J/m³ :). THE FIRST TWO ARE CONVENIENT APPROXIMATIONS

- NOTE THE ENERGY DENSITY DOES OSCILLATE WITH THE WAVE. USUALLY WE ONLY CONSIDER THE TIME-AVERAGE THOUGH.

POWER

THE RATE AT WHICH ENERGY IS DELIVERED (USUALLY AVERAGED)

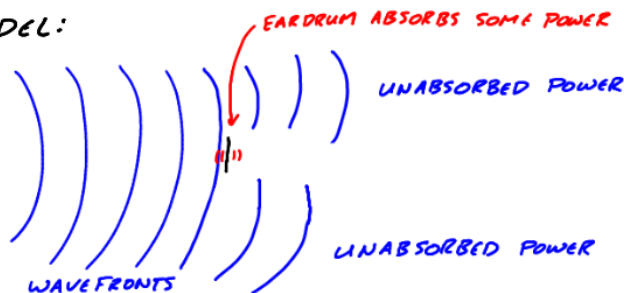
$$\text{POWER} = P = \frac{\text{ENERGY}}{\text{TIME}}, \text{ UNITS } \frac{\text{J}}{\text{s}} \equiv \text{W (WATT)}$$

"PUFFY P" TO DISTINGUISH FROM "PRESSURE"

- FOR MOST THINGS, 1W IS NOT A LOT.
 - O.G. LIGHTBULBS ABSORB 100W ELECTRICAL AND EMIT ~5W VISIBLE LIGHT
 - YOU GENERATE ~100W AT REST (WHY YOU NEED FUEL!)
 - TYPICAL HOME ~1000W AVERAGE
 - 1 HORSEPOWER = 745.7 W (CARS ~100 HP ~100 HOMES?)
 - FOR SOUND, 1W IS QUITE A LOT!
 - BREATHING QUIETLY ~ 10^{-10} W (100pW)
 - CONVERSATION ~ 10^{-6} W (10mW)
 - TRUMPET BLARING ~ 1W
- OBVIOUSLY DISTANCE PLAYS A ROLE, SO "POWER" IS NOT QUITE ENOUGH INFO TO TALK ABOUT "LOUDNESS".
- DON'T PUT YOUR EAR AT THE SOURCE!
- OUR PERCEPTION OF LOUDNESS MUST DEPEND PRIMARILY ON THE AMPLITUDE OUR EARDRUM OSCILLATES DUE TO A SOUND WAVE.

- MOTIVATION FOR INTENSITY:

- TOY MODEL:



10pt

Q

IF THE EARDRUM HAS HALF THE AREA, IT WILL ABSORB _____ THE POWER.

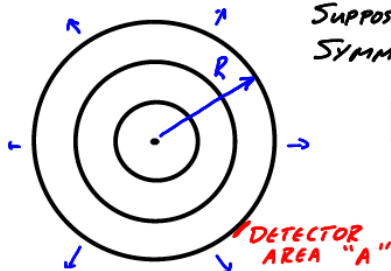
INTENSITY

THE POWER DELIVERED PER UNIT AREA

$$\text{INTENSITY} = I = \frac{\text{POWER}}{\text{AREA}}, \text{ UNITS } \frac{\text{W}}{\text{m}^2} = \frac{\text{J}}{\text{s m}^2} = \frac{\text{kg m}^2/\text{s}^2}{\text{s m}^2} = \frac{\text{kg}}{\text{s}^3}$$

- HOW DOES INTENSITY DEPEND ON DISTANCE FROM A SOURCE?

SUPPOSE A SOURCE EMITS 1W OF SOUND POWER, WHICH RADIATES SYMMETRICALLY IN ALL DIRECTIONS AT SPEED C.



10pt

Q

IF I PUT A SPHERICAL, SOUND-ABSORBING SHELL OF RADIUS "R" AROUND THE SOURCE, HOW MUCH POWER WILL IT ABSORB?

2026
Q IF I PUT A DETECTOR OF AREA "A" A DISTANCE R FROM THE SOURCE, HOW MUCH POWER WILL IT ABSORB?

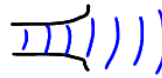
- SPREADING THE SOURCE POWER OVER $4\pi R^2$,

$\Rightarrow$ $INTENSITY = I = \frac{P}{4\pi R^2}$ TOTAL POWER
INTENSITY FALL-OFF WITH DISTANCE FROM SOURCE (SYMMETRIC EMISSION)
DISTANCE FROM SOURCE

- THIS IS A NICE QUANTITY BECAUSE IT'S INDEPENDENT OF WHAT DETECTOR WE USE! (SMALL OR BIG DETECTOR MEASURES THE SAME INTENSITY)

- CAVEATS ON $\frac{1}{R^2}$ SCALING:

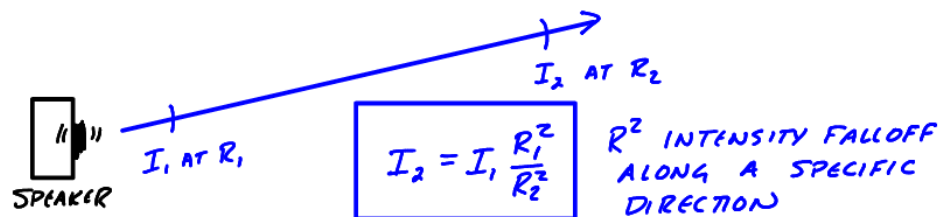
- ONLY SENSIBLE WHEN $R \gg$ SIZE OF THE SOURCE.

 1cm FROM A TUBA BELL, 1cm MAKES LITTLE DIFFERENCE! (CERTAINLY NOT A FACTOR OF 4!)

- MOST SOUNDS ARE AT LEAST SOMEWHAT DIRECTIONAL

- FOR LARGE R, STILL FALLS OFF LIKE $\frac{1}{R^2}$ IN A GIVEN DIRECTION, BUT EACH DIRECTION HAS A DIFFERENT PREFACTOR. (MORE LATER)

$\Rightarrow$ USUALLY PAY ATTENTION TO RELATIVE INTENSITY e.g.



AFTER MON
SEPT 23

- INTUITION:

IF SOMETHING EMITS INTO A CONE IT STILL SPREADS POWER OUT OVER A GROWING AREA:



RELATIONSHIP TO AMPLITUDE

- CONSIDER AIR'S QUANTITIES

ρ DENSITY (kg/m^3)

c SPEED OF SOUND (m/s)

v OSCILLATORY VELOCITY (m/s)

ΔP OSCILLATORY PRESSURE ($P_a = \text{N/m}^2 = \text{kg/ms}^2$)

I INTENSITY OF SOUND ($\text{W/m}^2 = \text{J/sm}^2 = \text{N/sm} = \text{kg/s}^3$)

20pt

Q THINKING BACK TO $E_{\text{kinetic}} = \frac{1}{2} m v^2$, GUESS

$$I \propto$$

20pt

Q JUST BY UNIT ANALYSIS, GUESS I IN TERMS OF ΔP .

$$I =$$

- IF ONE IS CAREFUL ABOUT PHYSICS,

BAR MEANS "TIME AVERAGED" (AFTER SQUARING)

NOTE FOR
POWER + INTENSITY
IF THE BAR IS
MISSING, ASSUME
IT'S THE AVERAGE
VALUE UNLESS
SPECIFIED AS
"INSTANTANEOUS"
OR "TIME-DEPENDENT"

$$\bar{I} =$$

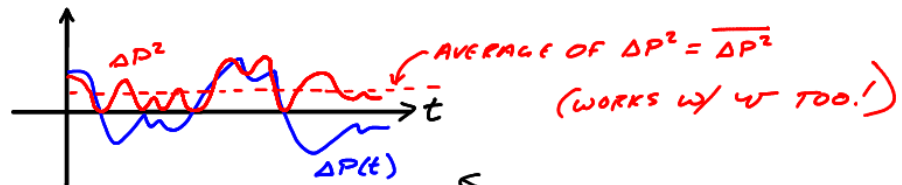
INTENSITY OF SOUND

NOTE: PRESSURE $\sim$ DENSITY WHICH IS WHY THERE IS THAT EXTRA ρ IN THE DENOMINATOR

c ON BOTTOM IS ALSO UNITATIVE?
HAS TO DO WITH THE WAY
 v DEPENDS ON c FOR A
GIVEN AMPLITUDE ΔP .

BONUS: DO THIS
AND FIND INTUITION!

- HOW TO CALCULATE $\overline{\Delta P^2}$:



- BUT REMEMBER: ALL SUCH WAVEFORMS CAN BE WRITTEN AS A
SUPERPOSITION OF PURE TONES!

SUPERPOSITIONS + INTENSITY SPECTRUM

10pt

Q WHAT IS THE INTENSITY OF $\Delta P = \Delta P_1 \cos(2\pi f t)$? (RECALL $I = \frac{\overline{\Delta P^2}}{\rho c}$)

30pt



HOW ABOUT TWO TONES? $\Delta P = \Delta P_1 \cos(2\pi f_1 t) + \Delta P_2 \cos(2\pi f_2 t)$

- EXTENDING THIS TO MORE PURE TONES,

$$\Rightarrow \boxed{\bar{I} = \bar{I}_1 + \bar{I}_2 + \bar{I}_3 + \dots}$$

↑ ↑ ↑
PURE TONE INTENSITIES (ALL FREQUENCIES MUST BE UNIQUE!)

SUMMATION RULE FOR PURE TONE INTENSITIES (OR POWERS)

- ADDED NEATNESS: REMEMBER FOURIER SERIES?

$$\Delta P(t) = \Delta P_1 \cos(2\pi f_1 t + \phi_1) + \Delta P_2 \cos(2\pi f_2 t + \phi_2) + \dots$$

$$\Rightarrow \bar{I} = \bar{I}_1 + \bar{I}_2 + \dots$$

$$= \frac{\Delta P_1^2}{2\rho c} + \frac{\Delta P_2^2}{2\rho c} + \dots$$

EACH $\bar{I}_n$ IS THE INTENSITY AT FREQUENCY f_n , +

$\bar{I}_n(f_n)$ IS THE "INTENSITY SPECTRUM" YOU SEE IN AUDIO EDITING.

"INTENSITY SPECTRUM" →



- SUMMING THE INTENSITY SPECTRUM GIVES THE TOTAL INTENSITY!

- SUMMING A FEW NEIGHBORING VALUES GIVES THE INTENSITY FOR THAT "BAND." ☺

ANYWAY:

- THE BEAUTY OF DEFINITIONS LIKE THESE ARE THAT EVERYONE ^(COMPETENT) WILL MEASURE THE SAME INTENSITY, WHILE "LOUDNESS" IS PERCEIVED DIFFERENTLY BY DIFFERENT EARS & BRAINS. (4 SITUATIONS)

- BUT HOW DOES INTENSITY RELATE TO PERCEIVED LOUDNESS?

- IS THERE A "NATURAL" SCALE FOR LOUDNESS?

Q CHARACTERIZE THE STEADINESS OF LOUDNESS CHANGE FOR EACH PLAYED AUDIO TONE.

DRAW QUALITATIVE LOUDNESS VS t TO HELP?

EXPECTED?

- a)
- b)
- c)
- d)
- e)
- f)

- (1) SPEEDS UP A LOT
- (2) SPEEDS UP A BIT
- (3) STEADILY CHANGES
- (4) SLOWS DOWN A BIT
- (5) SLOWS DOWN A LOT

INCLUDE RZ FALLOFF (TRAIN)

DEMO

PLAY FILES a-f (BEFORE FILLING IN RED OR BLUE!)

COMMON ANSWERS (WE ARE BIASED BY MANY THINGS, INCLUDING THE ORDER IN WHICH THEY'RE PLAYED)

DEMO

- SHOW THE PLOTS OF a, b, c ON A LINEAR SCALE, ZOOM AROUND (THIS CAN BE DONE BY RUNNING "RAMPED LOUDNESS.py")
- DISCUSS LOG SCALE + RELATE TO PERCEPTION
- PLAY SOUNDS AGAIN. (NOTE ANY CHANGES DUE TO TRAINING BIAS?)

SUBJECTIVE! DEPENDS ON MAGNITUDE!
THIS IS ONLY CONVENTIONAL WISDOM.

IF WE HEAR INTENSITY ON A LOG-ISH SCALE,
IS THIS NOT SURPRISING?

- THIS IS HOW THE AMPLITUDES OF VIBRATIONS NATURALLY DECAY (AND NOT JUST TUNING FORKS...)

DEEP REASON: LOSS ^{OR GROWTH} RATES OF QUANTITIES & QUANTITIES

$$\text{EVERYWHERE IN NATURE!!} \quad \left\{ \begin{array}{l} \Rightarrow \frac{dQ}{dt} = -kQ \\ \Rightarrow Q \propto e^{-kt} \end{array} \right.$$

→ SOUND, ELECTROMAGNETISM, NUCLEI, ATOMS, POPCORN, POPULATIONS, ...

- OUR EARS CAN SENSE INTENSITIES SPANNING MANY ORDERS OF MAGNITUDE! (MORE LATER)
(MORE IMPORTANT FOR HANDLING EVERY SITUATION?)

END OF WED
SEPT 25

⇒ IT IS CONVENIENT TO USE A LOG SCALE

DECIBEL SCALE

- WE HAVE DABBLED IN LOG SCALES ALREADY
- IN ORDER TO TALK TO AUDIO ENGINEERS ABOUT LOUDNESS, WE NEED TO KNOW THE DECIBEL SCALE.

REMEMBER FREQUENCIES,
EARS, MUSICAL SCALES?

DEFINITION: 10 DECIBELS (dB) = A FACTOR OF 10 IN POWER
= A FACTOR OF $\sqrt{10}$ IN AMPLITUDE

DECIBEL
SCALE,
SIMPLE
DEFINITION

MOTIVATION:
MUCH EASIER
TO SAY / PLOT
THAN, E.G.
SCIENTIFIC
NOTATION
OR S.I.
PREFIXES

dB	POWER RATIO	AMPLITUDE RATIO
0	1	1
10pt +10	10	$\sqrt{10} \approx 3.16$
10pt [Q]	100	
+30	1000	
+40	10,000	$\sqrt{1000} \approx 31.6$
10pt :	:	100
10pt [Q]	:	:
10pt [Q] +100	:	:
-10		
-20	0.01	0.1
:	:	:
-100	0.0000000001	0.00001

FUN:
→ "BEL" IS A
UNIT NAMED
AFTER ALEXANDER
GRAHAM BELL.
→ 1 BEL = $\times 10$
→ EARTHQUAKE
MAGNITUDES...

- MATH DEFINITION:

SAME FOR INTENSITIES!

$$\text{POWER} = (\text{REFERENCE POWER}) 10^{(\text{DIFFERENCE IN dB})/10}$$

$$\text{DIFFERENCE IN dB} = 10 \log_{10} \left(\frac{\text{POWER}}{\text{REFERENCE POWER}} \right)$$

$$P_2 = P_1 10^{(dB_2 - dB_1)/10}$$

$$dB_2 - dB_1 = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

- NOTE: DECIBELS MAKE STATEMENTS ABOUT RELATIVE VALUES, NOT ABSOLUTE!

VAGUE PERSON: "THAT'S 23 dB"

PRECISE PERSON: "23dB RELATIVE TO WHAT?"

10pt

Q IF 3dB IS A FACTOR OF ≈ 2 (IT IS VERY CLOSE!)
9dB IS A FACTOR OF

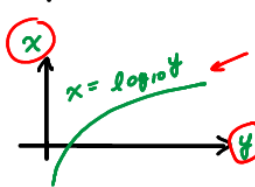
REVIEW OF LOGARITHMS

- LOGARITHMS "UNDO" EXPONENTIALS:

$$y = 10^x$$

LOG BOTH SIDES $\Rightarrow \log_{10}(y) = \log_{10}(10^x)$

SIMPLIFY $\Rightarrow \log_{10} y = x$



"FLIPPED" OR "INVERSE" OF EXPONENTIAL

- FUN WITH LOGS:

$$\log_{10}(10^a 10^b) = \log_{10}(10^{a+b})$$

$$\log_{10}(10^a / 10^b) = \log_{10}(10^{a-b})$$

$$\log_{10}((10^a)^c) = \log_{10}(10^{ca})$$

$$\text{DEFINE } A \equiv 10^a \text{ OR } a = \log_{10} A$$

$$B \equiv 10^b \text{ OR } b = \log_{10} B$$

$$\Rightarrow \log_{10}(AB) = a + b = \log_{10} A + \log_{10} B$$

$$\Rightarrow \log_{10}(A/B) = a - b = \log_{10} A - \log_{10} B$$

$$\Rightarrow \log_{10}(A^c) = ca = c \log A$$

10pt

Q WHAT IS THE DIFFERENCE IN dB BETWEEN INTENSITIES I_1 & I_2 ?

10pt

Q WHAT IS THE DIFFERENCE IN dB BETWEEN WAVES OF AMPLITUDES A_1 & A_2 ?

20pt

Q

SOMEONE STANDING A DISTANCE R_1 FROM A SOUND SOURCE MEASURES AN INTENSITY I_1 . HOW MANY DECIBELS DIFFERENCE WILL SOMEONE AT DISTANCE $R_2 = 2R_1$ MEASURE?

EAR

- AMAZING DETECTOR WITH FANTASIC "DYNAMIC RANGE"

- TRULY "READY FOR EVERYTHING"

THRESHOLD OF HEARING
WHISPER
NORMAL CONVERSATION
BUSY STREET TRAFFIC
THRESHOLD OF PERMANENT DAMAGE
FRONT ROW OF ROCK CONCERT
PAIN THRESHOLD
INSTANT PERFORATION OF EAR DRUM

LEAVE
UP

TOTAL
POWERS

RANGE OF ACCEPTABLE INPUTS

"DYNAMIC RANGE"

16 ORDERS OF
MAGNITUDE!!

$10^{-12} \text{ W/m}^2 = 1 \text{ pW/m}^2$
 $10^{-10} \text{ W/m}^2 = 100 \text{ pW/m}^2$
 $10^{-6} \text{ W/m}^2 = 1 \text{ mW/m}^2$
 $10^{-5} \text{ W/m}^2 = 10 \text{ mW/m}^2$
 $10^{-3} \text{ W/m}^2 = 1 \text{ W/m}^2$
 $10^{-1} \text{ W/m}^2 = 0.1 \text{ W/m}^2$
 $10^1 \text{ W/m}^2 = 10 \text{ W/m}^2$
 $10^4 \text{ W/m}^2 = 10,000 \text{ W/m}^2$

- 16 ORDERS: HEAR EVERYTHING FROM
 ~ 1 TO $\sim 10,000,000,000,000,000$

16 ZEROS

OR 160dB DYNAMIC RANGE

(THOUGH ONE COULD ARGUE ABOUT THE WORD "ACCEPTABLE")

EAR DRUM IS $\sim 5 \times 10^{-5} \text{ m}^2$
 SO A WATT IS ABSRD
 TO "HEAR"!

TECHNOLOGY	AUDIO BITS	DYNAMIC RANGE
NINTENDO (NES)	8	$(2^8)^2 = 48 \text{ dB}$
CDs	16	$(2^{16})^2 = 96 \text{ dB}$
FANCY AUDIO	24	$(2^{24})^2 = 144 \text{ dB}$
OVERKILL AUDIO	32	$(2^{32})^2 = 192 \text{ dB}$
BEST AUDIO AMPS I FOUND		$\sim 120 \text{ dB}$

NOTE I NEED TO
CHECK IF THIS #
IS WHAT I THINK
IT IS...

THIS LIMITS EVERYTHING, AND IS WHY YOU CAN HEAR
THE HISS WHEN YOU TURN ANYTHING UP HIGH ENOUGH
TO HURT YOUR EARS.

⇒ GET A COMPRESSOR + NOISE GATE FOR A CONCERT?

- DECIBELS ARE A RELATIVE SCALE, SO WHAT ABOUT AN ABSOLUTE
LOG SCALE?

END OF
MON SEPT 30

SOUND INTENSITY LEVEL (SIL)

- SINCE THE THRESHOLD OF HEARING IS $\sim 1 \text{ pW/m}^2$, LET'S USE " 1 pW/m^2 " AS THE REFERENCE:

$$\text{SIL [dBp]} = 10 \log_{10} \left(\frac{\text{INTENSITY}}{1 \text{ pW/m}^2} \right)$$

SOUND INTENSITY
LEVEL. (SOMETIMES
"SOUND PRESSURE
LEVEL" OR SPL)

UNITS "dB RELATIVE TO pW/m^2 "

THE "p" IS NOT STANDARD
AND CAN BE OMITTED. I WILL
LEAVE IT HERE TO REMIND US THAT
THIS IS AN ABSOLUTE SCALE

(LIKE "dBm" IN MICROWAVE ENGINEERING)
↑ "MILLIWATT"

10pt

Q CALCULATE SIL FOR

THRESHOLD OF HEARING
THRESHOLD OF PAIN
INSTANT PERFORATION

10pt

Q ONE SOUND HAS $\text{SIL} = 97 \text{ dBp}$, ANOTHER HAS 100 dBp .
BY WHAT FACTOR IS THE SECOND SOUND'S PRESSURE AMPLITUDE
LARGER?

LOUDNESS

"LOUDNESS" IS SUBJECTIVE, BUT PERCEIVED TO INCREASE WITH POWER, AMPLITUDE, OR SIL, PERHAPS MORE CLOSELY FOLLOWING A $\log(\text{dB})$ SCALE (OVER SOME RANGE)

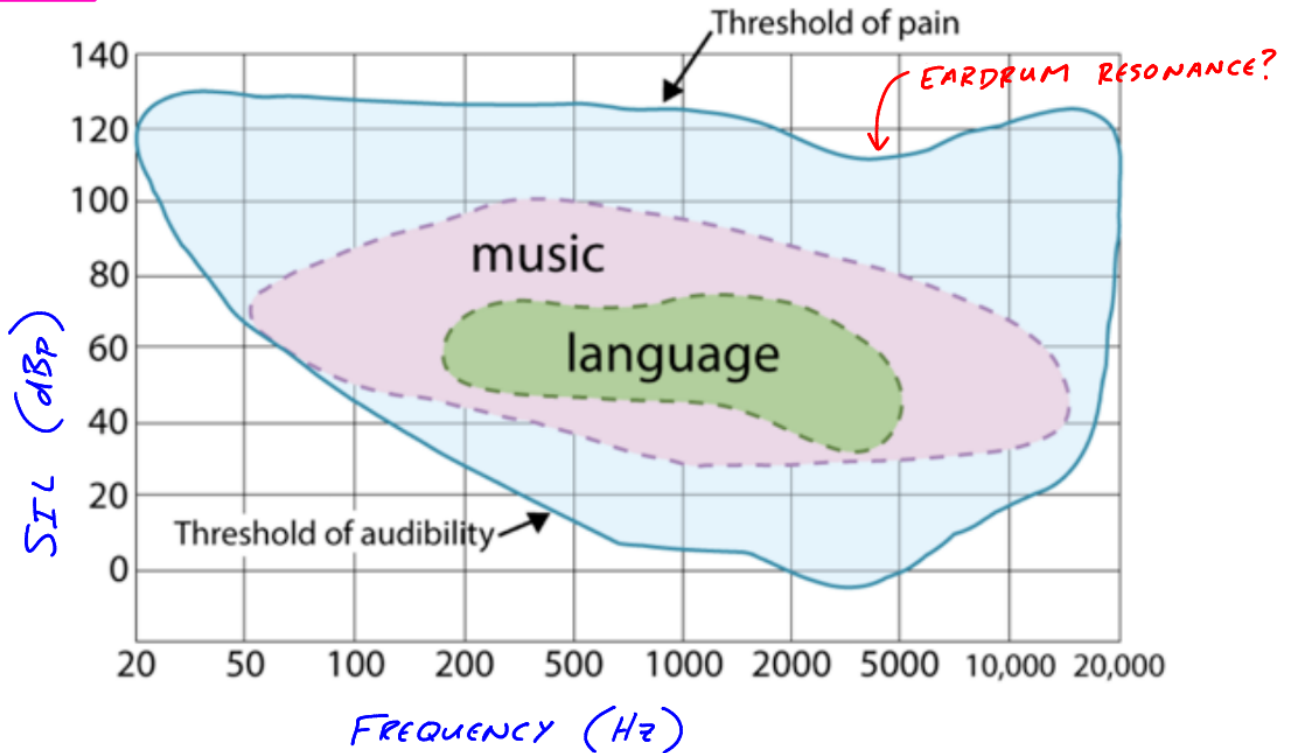
HOWEVER, THIS PERCEPTION IS COMPLEX, AND, e.g. 60 dBp DOES NOT SOUND "TWICE AS LOUD" AS 30 dBp !

(BUT NOW WE HAVE THE TOOLS TO QUANTIFY THIS)

WARNING:

- DAMAGE TO THE SENSORS UNDER YOUR "EAR BARS" OCCURS ABOVE THE "SAFE" SIL OF 85dBp.
- YOUR "LOG-ISH" PERCEPTION CAN HIDE BIG CHANGES IN POWER (WHICH IS A BETTER FIGURE OF MERIT FOR DAMAGE).

DEMO RANGE OF HEARING PLOT



SUMMARY

$\cos^2$ AND $\sin^2$ AVERAGE TO $\frac{1}{2}$

KEEP THIS IN YOUR POCKET OF TRICKS

ENERGY & WORK

$\Delta E = F \Delta x$ WORK DONE BY A FORCE.
 CONSTANT FORCE
 DISTANCE PUSHED

$E_{\text{KINETIC}} = \frac{1}{2} m v^2$ KINETIC ENERGY, UNITS $Nm = J$
 HEAVIER, FASTER $\Rightarrow$ MORE KINETIC ENERGY
 DOES NOT DEPEND ON DIRECTION
 CONSTANT $\approx 9.8 \text{ m/s}^2$
 "JOULE"

$mg \Delta x = U_{\text{GRAVITY}}$ GRAVITATIONAL POTENTIAL (EARTH'S SURFACE)
 HEIGHT

$U_{\text{SPRING}} = \frac{1}{2} k \Delta x^2$ SPRING POTENTIAL ENERGY
 SPRING CONSTANT (N/m^2)
 DISTANCE COMPRESSED OR EXTENDED

WAVE ENERGY, POWER, INTENSITY

WAVE ENERGY DENSITY $\propto (\text{AMPLITUDE})^2$
 COULD BE PRESSURE, VELOCITY, DISPLACEMENT, DENSITY, ...

POWER = $\frac{\text{ENERGY}}{\text{TIME}} = P$ UNITS $J/s \equiv W$ (WATT)

INTENSITY = $\frac{\text{POWER}}{\text{AREA}} = I$ UNITS W/m^2

$I = \frac{P}{4\pi R^2}$ INTENSITY FALL-OFF WITH DISTANCE FROM SOURCE (SYMMETRIC EMISSION)
 TOTAL POWER OUTPUT FROM SOURCE
 DISTANCE FROM SOURCE

$\Rightarrow I_2 = I_1 \frac{R_1^2}{R_2^2}$ RELATIVE INTENSITY AT TWO DISTANCES R_1, R_2

$I = \rho \bar{v}^2 c$
 $= \frac{\Delta p^2}{\rho c}$ INTENSITY FOR SOUND WAVES FOR VELOCITY AMPLITUDE v OR PRESSURE AMPLITUDE Δp

$I = I_1 + I_2 + I_3 + \dots$ SUMMATION RULE FOR PURE TONE INTENSITIES (OR POWERS)
 PURE TONE INTENSITIES (ALL FREQUENCIES MUST BE UNIQUE!)
 $I_n = \frac{\Delta p_n^2}{2\rho c}$ AMPLITUDE OF PURE TONE. FROM $\cos^2()$

DECIBEL SCALE

10 DECIBELS (dB) = A FACTOR OF 10 IN POWER (OR INTENSITY)
 = A FACTOR OF $\sqrt{10}$ IN AMPLITUDE

$\Rightarrow \text{DIFFERENCE IN dB} = 10 \log_{10} \left(\frac{\text{POWER}_1}{\text{POWER}_2} \right)$
 $\Rightarrow \frac{\text{POWER}_1}{\text{POWER}_2} = 10^{(\text{DIFFERENCE IN dB})/10}$ DECIBEL SCALE MATH.

$SIL [dBp] = 10 \log_{10} \left(\frac{\text{INTENSITY}}{1 \text{ pW/m}^2} \right)$ SOUND INTENSITY LEVEL (RELATIVE TO pW/m^2 THRESHOLD)
 $0 \text{ dBp} \rightarrow 1 \text{ pW/m}^2$

ASSUME THESE ARE TIME-AVERAGED UNLESS SPECIFIED!